# Crack IQI and Shim Procurement Specifications

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[redacted]

NASA Johnson Space Center

Rev. A

# Crack IQI Procurement Specification

- Objective: Develop and validate crack IQI design and manufacturing with goal of standardizing crack IQI specification and use

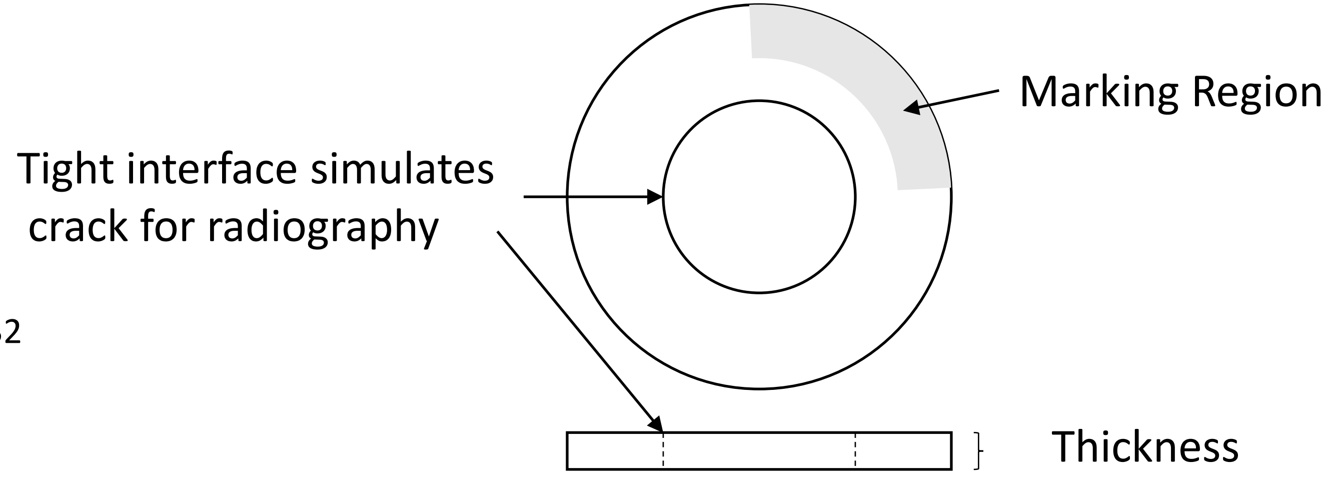
Notes:
1. Material: Annealed Ti 6Al-4V Bar per AMS 4967
2. Number of  Cylinder-shaft assemblies: 1 each
    1. One for EB weld
3. Crack IQI Thicknesses from each assembly and Marking
    1. 0.030” (3 each), Etch Marking: CIQI-Ti-0.030
    2. 0.050” (3 each), Etch Marking: CIQI-Ti-0.050
    3. 0.100” (3 each), Etch Marking: CIQI-Ti-0.100
    4. 0.200” (3 each), Etch Marking: CIQI-Ti-0.200
4. Grind Crack IQI faces to surface finish of Ra 0.8 micrometer (32 microinch)
5. Deburr OD corners
6. Deliver left over cylinder-shaft assembly, 1 each
7. Both parts made from the same piece of bar stock
8. FN 1 shrink fit tolerances on the cylinder hole and the shaft per ANSI B4.1
9. Hole interior and shaft exterior surface roughness desired (√): Ra 0.1 to 0.4 micrometers (µm)
10. Crack IQI interference average gap desired: ~2 micrometers
    1. Measured by NASA or if possible, by vendor

11.Dimensions in inches. Surface finish in micrometers.

12.Tolerance: X.X = +/

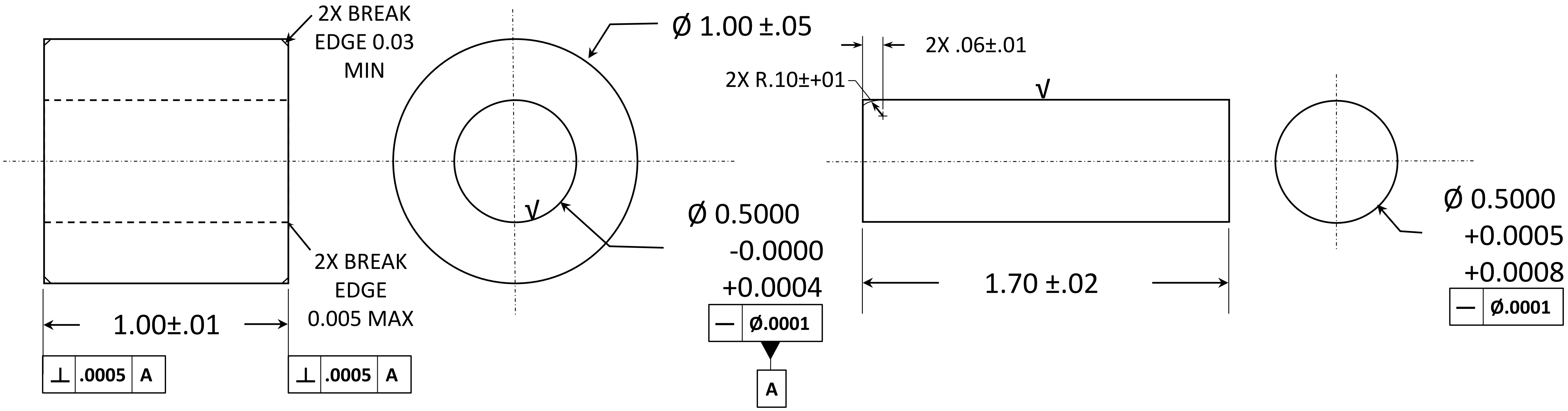
13.0.1, X.XX = +/ 0.01. X.XXX =+/ 0.002 Unless

14. Provide a certificate conformance to above requirements




Crack IQI Schematic

# Crack IQI Parts




Crack IQI Cylinder

Crack IQI Shaft

# ANSI B4.1

Force or Shrink Fits Tolerances per. ANSI B4.1 Table Calculator

Variables

| | |
|---|---|
| Enter the desired tolerance = | FN 1 |
| Enter shaft nominal diameter (inch) = | 0.5000 |

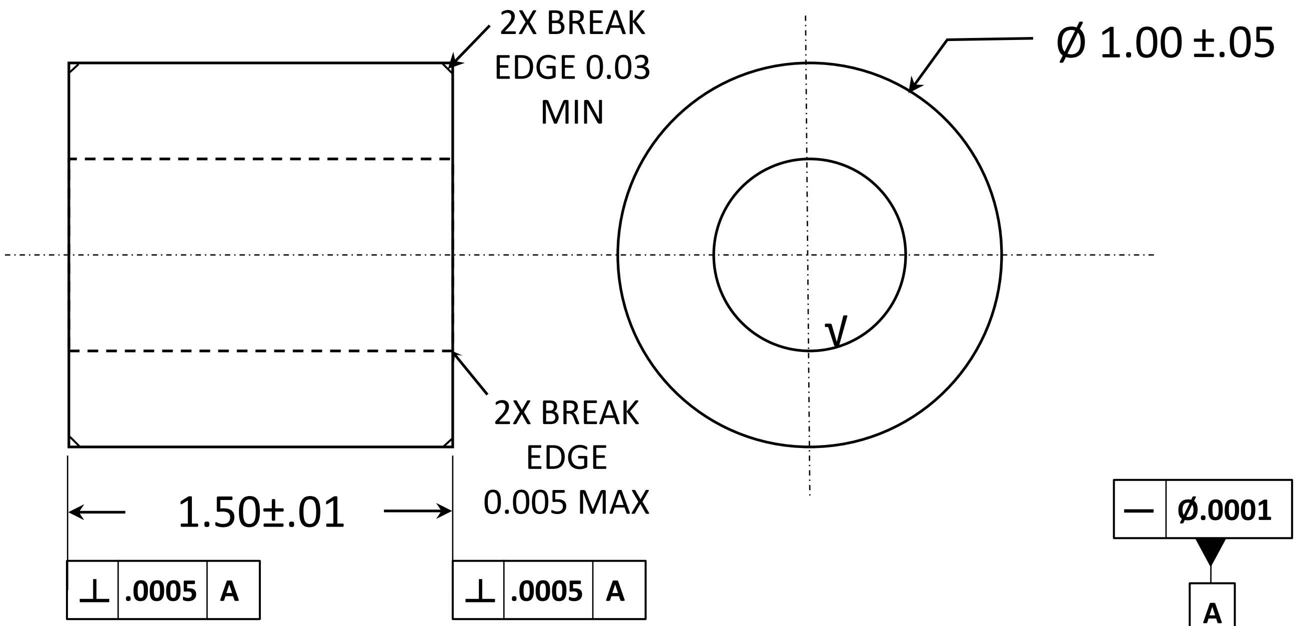
Results

| | Max | Min |
|---|---|---|
| Tolerance (thousandths in) = | 0.8 | 0.5 |
| SHAFT Diameter max / min in. = | 0.50080 | 0.50050 |
| Tolerance Zone = | - | |

| | Max | Min |
|---|---|---|
| Tolerance (thousandths in) = | 0.4 | 0 |
| HOLE Diameter max / min in. = | 0.50040 | 0.50000 |
| Tolerance Zone = | H6 | |

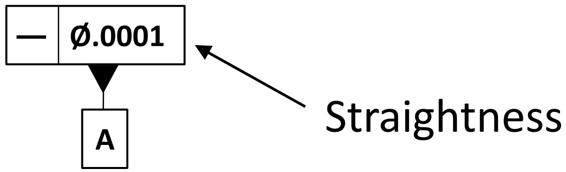
| | Max | Min |
|---|---|---|
| Interference (in) = | 0.00080 | 0.00010 |

# Cylinder

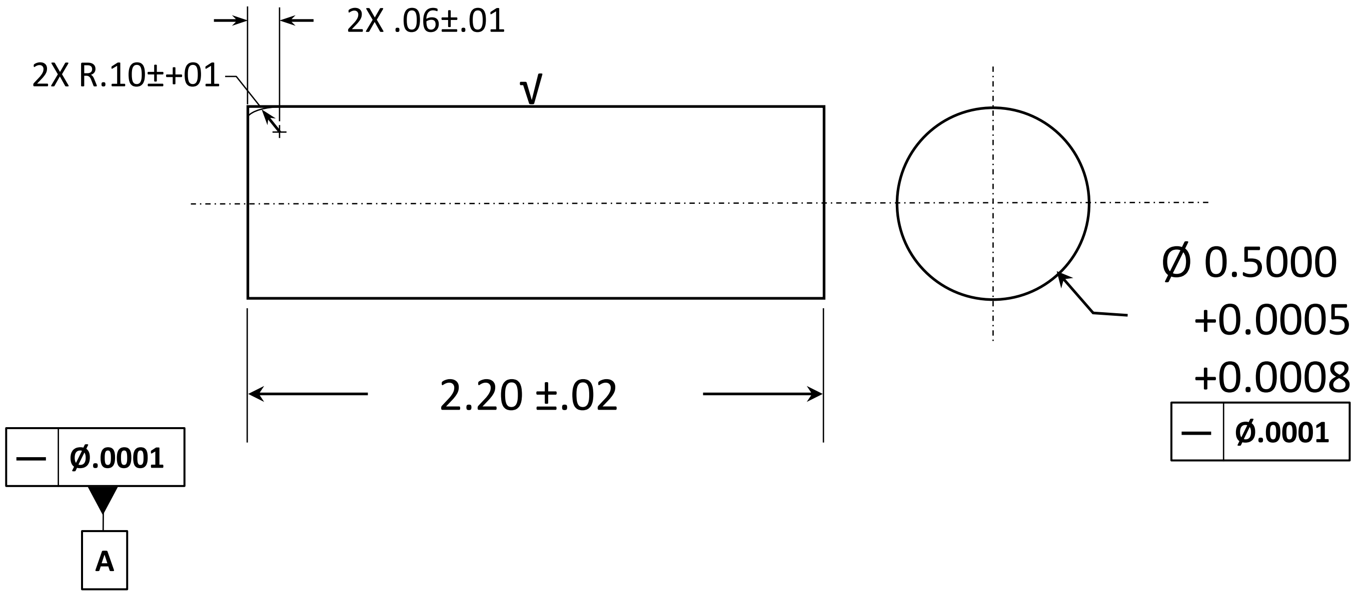



Crack IQI Cylinder

| | Max | Min |
|---|---|---|
| Tolerance (thousandths in) = | 0.4 | 0 |
| HOLE Diameter max / min in. = | 0.50040 | 0.50000 |
| Tolerance Zone = | H6 | |



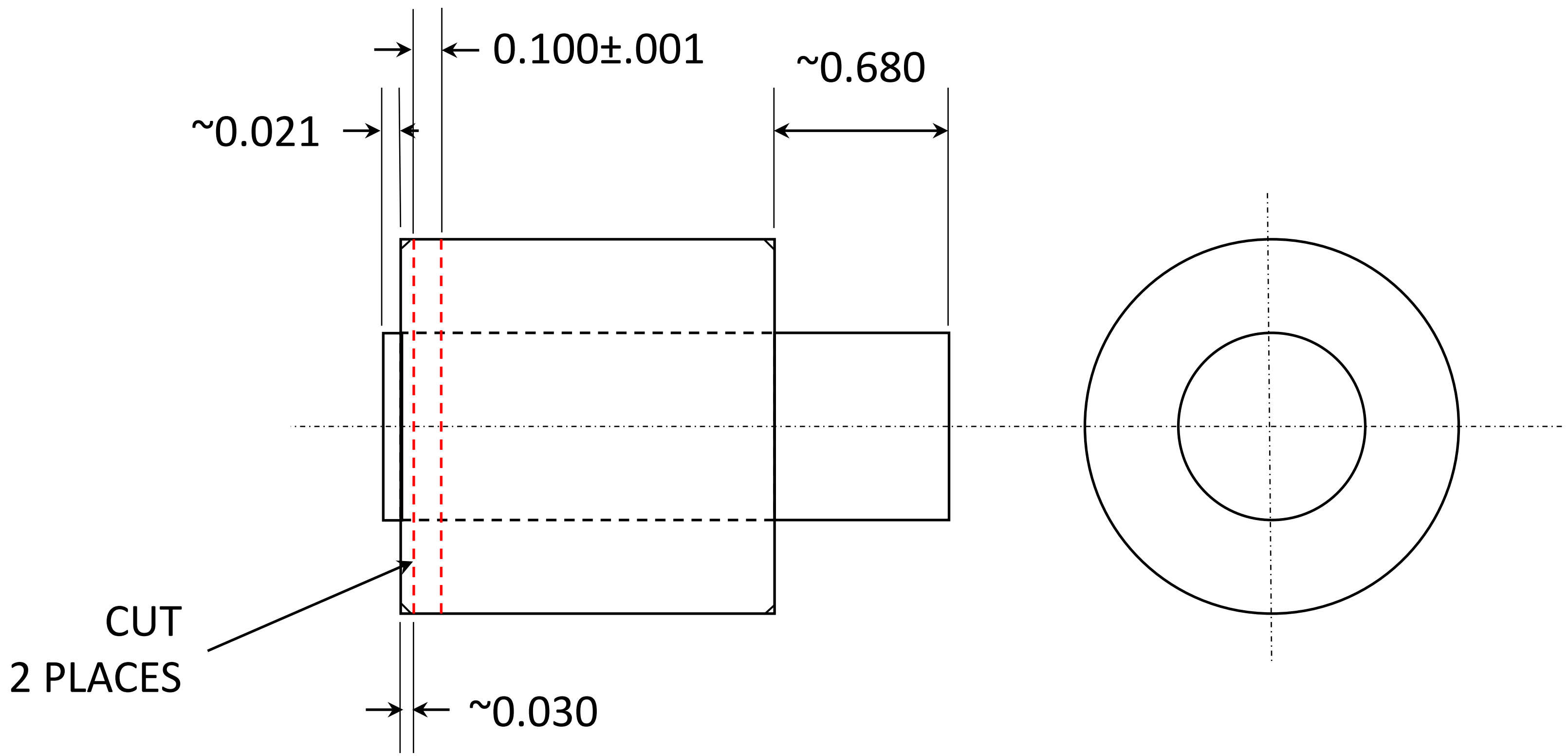
# Shaft



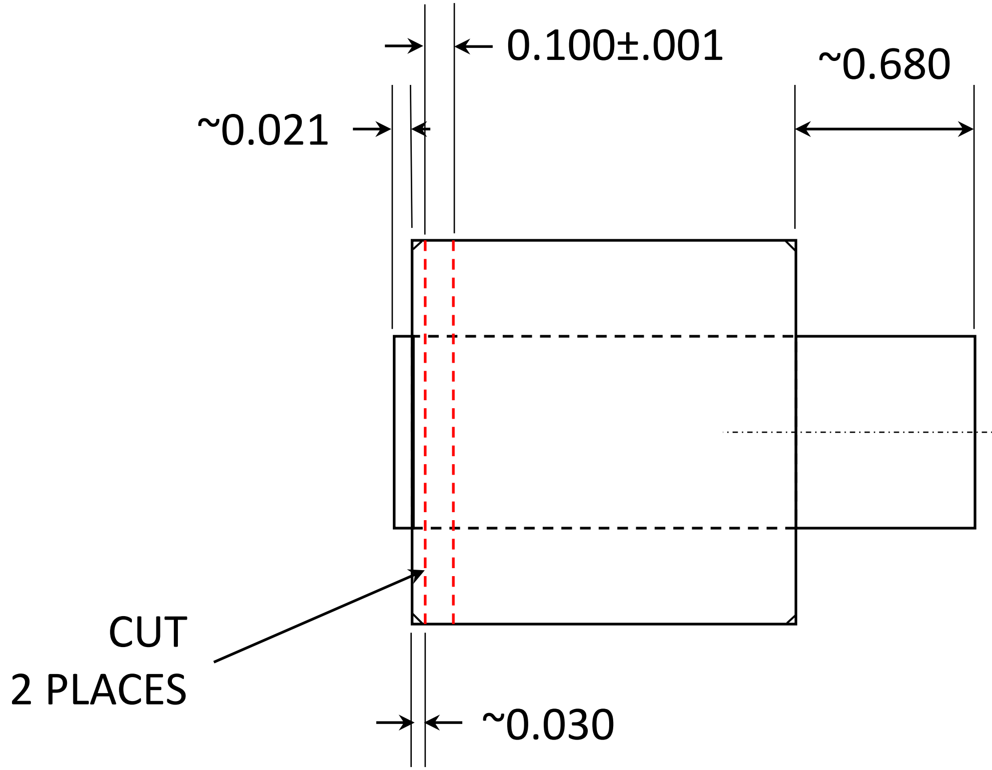

Crack IQI Shaft

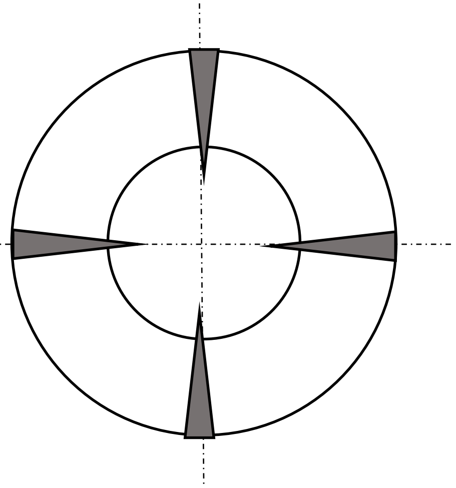
| | Max | Min |
|---|---|---|
| Tolerance (thousandths in) = | 0.8 | 0.5 |
| SHAFT Diameter max / min in. = | 0.50080 | 0.50050 |
| Tolerance Zone = | - | |

# Cylinder-shaft Assembly and Cut Plane



# Disc to Ring Weld Attachment for Each crack IQI Slice



EB Weld
(4 places, along entire length)

Notes for Disc to Ring Weld Attachment for Each crack IQI Slice

- Method: Electron Beam weld of Cylinder Shaft assembly before slicing individual Crack IQIs

# Shim Specifications

- Notes:

1. Material: Annealed Ti 6Al-4V Bar per AMS 4967
2. Shim Thicknesses and Marking

    1. 0.020” (1 each), Etch Marking: SHIM-Ti-0.020
    2. 0.033” (1 each), Etch Marking: SHIM-Ti-0.033
    3. 0.067” (1 each), Etch Marking: SHIM-Ti-0.067
    4. 0.133” (1 each), Etch Marking: SHIM-Ti-0.133

3. Grind/machine shim faces to surface finish of Ra 0.8 micrometer (32 microinch)
4. Deburr OD corners
5. Dimensions in inches. Surface finish in micrometers.
6. Tolerance: X.X = +/-0.1, X.XX = +/-0.01. X.XXX =+/-0.002 Unless
7. Provide a certificate conformance to above requirements